

17. Walking the DOM (DOM me ghumna)

Saare operations `document` se shuru hote hain. Isse kisi bhi node tak pahunch sakte ho.

Sabse upar

Element	Kaise milega
<code><html></code>	<code>document.documentElement</code>
<code><body></code>	<code>document.body</code>
<code><head></code>	<code>document.head</code>

Dhyan: agar script `<head>` me hai, to `document.body` `null` hoga (abhi body parse nahi hui). DOM me `null` = "exist nahi karta".

Terms

- **Children:** seedhe andar wale nodes.
- **Descendants:** saare nested nodes (children ke children bhi).

Sab nodes ke liye (text/comment bhi included)

Property	Kaam
<code>childNodes</code>	saare child nodes
<code>firstChild</code> , <code>lastChild</code>	pehla aur aakhri child
<code>parentNode</code>	parent
<code>previousSibling</code> , <code>nextSibling</code>	bagal wale nodes
<code>hasChildNodes()</code>	child hai ya nahi

Sirf Elements ke liye (text/comment skip)

Property	Kaam
<code>children</code>	sirf element children
<code>firstElementChild</code> , <code>lastElementChild</code>	
<code>previousElementSibling</code> , <code>nextElementSibling</code>	
<code>parentElement</code>	parent element

`parentNode` vs `parentElement`: sirf `document.documentElement` (`<html>`) me fark hai. Uska `parentNode` = `document`, par `parentElement` = `null`.

js

```
while (elem = elem.parentElement) { /* <html> tak upar jao */ }
```

DOM Collections

`childNodes` array nahi, ek collection hai (array-like, iterable).

- `for..of` chalta hai. `for..in` **mat** use karo (extra properties aati hain).
- Array methods (`filter` etc.) nahi chalte. `Array.from(collection)` se array bana lo.
- **Read-only** hain (`childNodes[i] = ...` nahi kar sakte).
- Zyaadatar **live** hote hain (DOM badle to apne aap update).

Tables ke special properties

- `table.rows`, `table.tBodies`, `table.caption`, `table.tHead`, `table.tFoot`
- `tbody.rows`
- `tr.cells`, `tr.rowIndex`, `tr.sectionRowIndex`
- `td.cellIndex`

js

```
let td = table.rows[0].cells[1]; // pehli row, doosra column  
td.style.backgroundColor = "red";
```

Tasks se seekh

- `elem.lastChild.nextSibling` hamesha `null` hota hai (sahi).
- `elem.children[0].previousSibling` hamesha `null` **nahi** hota, kyunki pehle text node ho sakta hai.

Yaad rakho

Sab nodes wale set: `parentNode`, `childNodes`, `firstChild`, `lastChild`, `previousSibling`, `nextSibling`. Sirf elements wale set: `parentElement`, `children`, `firstElementChild`, `lastElementChild`, `previousElementSibling`, `nextElementSibling`.